# Sampling distributions

**Topic 6 · Thursday, January 21 and Tuesday, January 26**

EDUC 15: Statistics for Education Research
UCI School of Education

# Where are we?

**Description**

Summarize data

Visualize data

**We are here, heading right**



**Prediction**

**Explanation**

Peter the Anteater at a UC Irvine basketball game.

# The question we keep asking

## If what I think about the population is true, how weird is my sample?

So far, a sample was one person. We asked how weird that one score was

A real sample has many people. So we compute its mean, and ask how weird that mean is

Then, how weird must it be before I change my mind? That's Topic 7

# Plan for the rest of today

**The distribution of sample means**

**How weird is one mean?**

**On Tuesday: standard errors, how to estimate them, and how to use them**

# The distribution of sample means

# Our blueprint: three panels, one story

## Panel 1: The population

Everyone we care about. Its μ and σ are fixed, and usually hidden from us

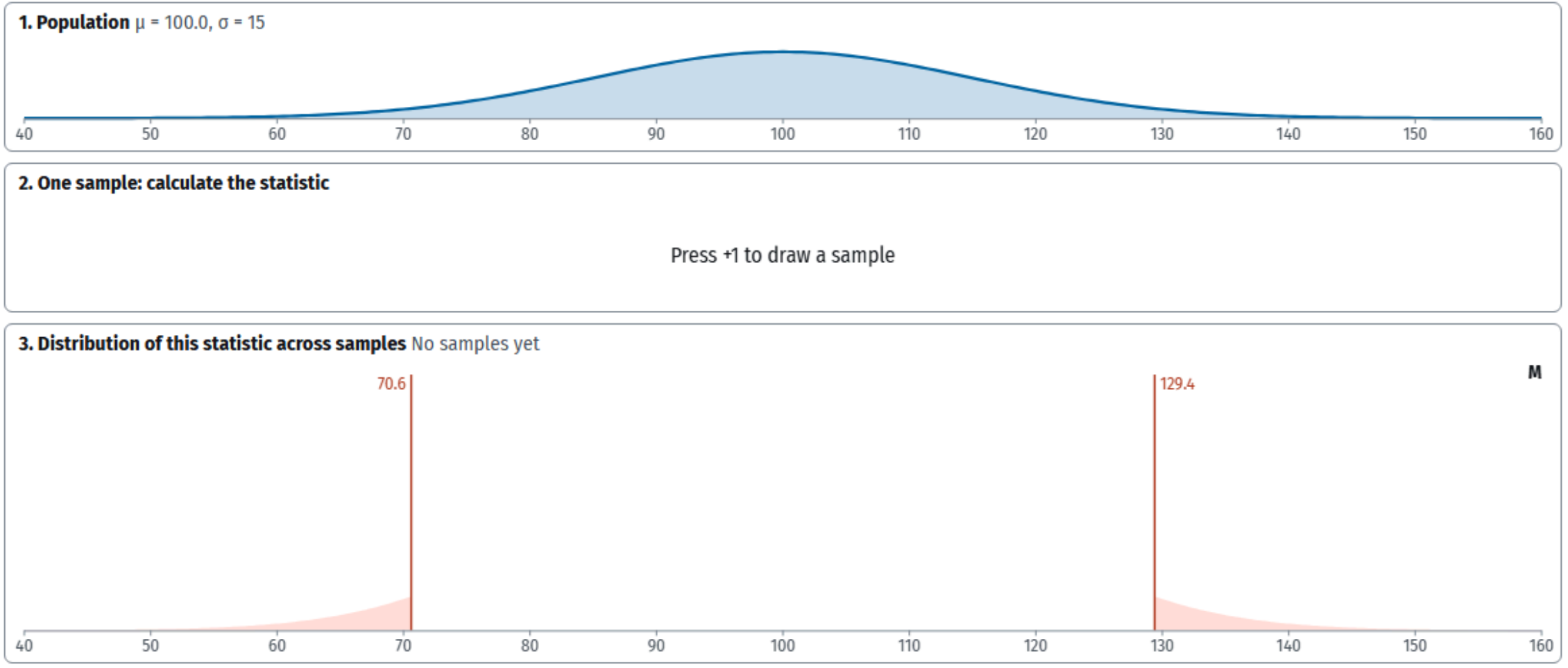
## Panel 2: One sample

We draw n people and compute one statistic, such as their mean M

## Panel 3: That statistic, across samples

We repeat, and watch how the statistics stack up

# Start small: samples of one person




Simulator after Kristoffer Magnusson, rpsychologist.com.

# That gap is called sampling error

## In points

$M - \mu = 106.2 - 100 = \mathbf{6.2}$

## In SDs: a z-score

$(106.2 - 100) / 15 = \mathbf{0.41}$

**Sampling error is the gap between a statistic and the parameter it estimates**

**Nobody made a mistake. A random sample rarely matches its population exactly**

# Predict it: Fifty more samples of one

**We have one dot in panel 3, at 106.2. We draw 50 more samples, then 500 more. Each gives one mean, and each mean is one dot**

**What will the 551 dots look like?**

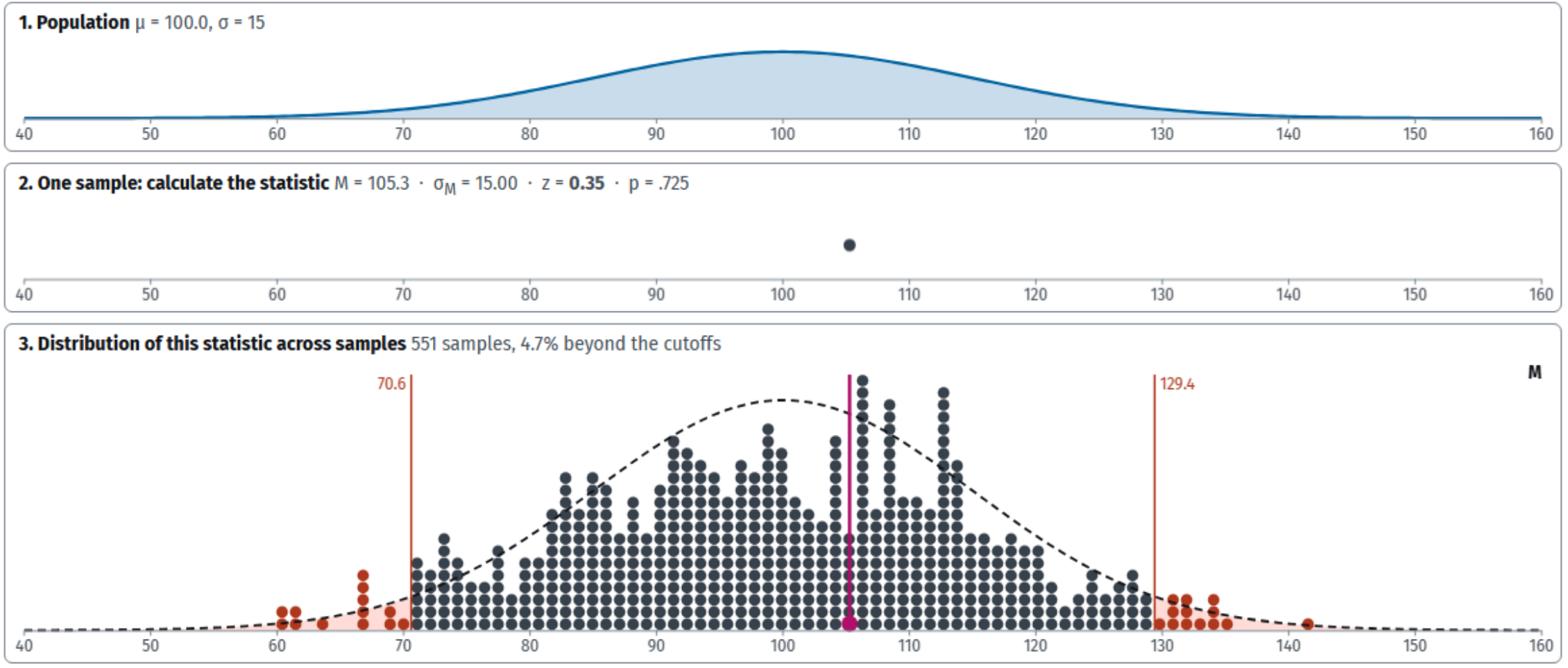
**A** A bell around 100, as wide as the population

**B** A narrow spike right at 100

**C** A flat line across the whole axis

**D** A bell around 106.2, our first sample

# n = 1: the means copy the population




Simulator after Kristoffer Magnusson, rpsychologist.com.

# How weird is one mean?

# That pile has a name

## Sampling distribution

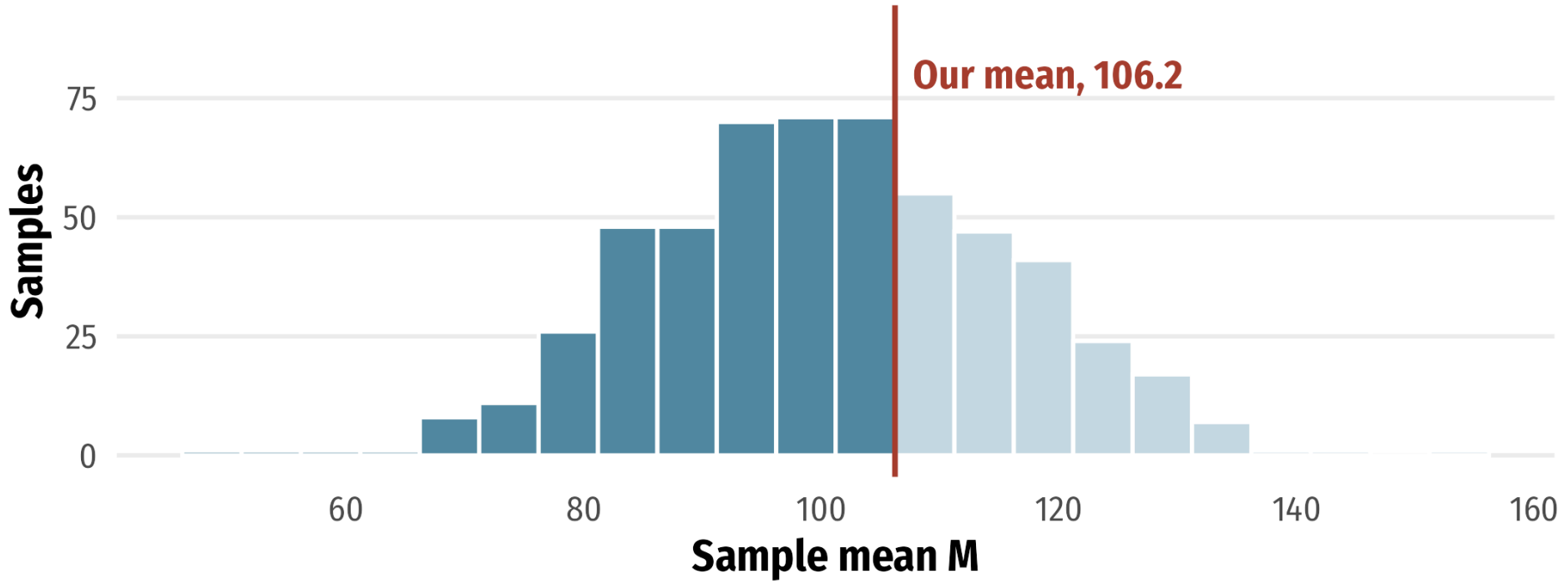
The distribution of a statistic, across all possible samples of size n

## Distribution of sample means

The same thing, when the statistic is the mean M

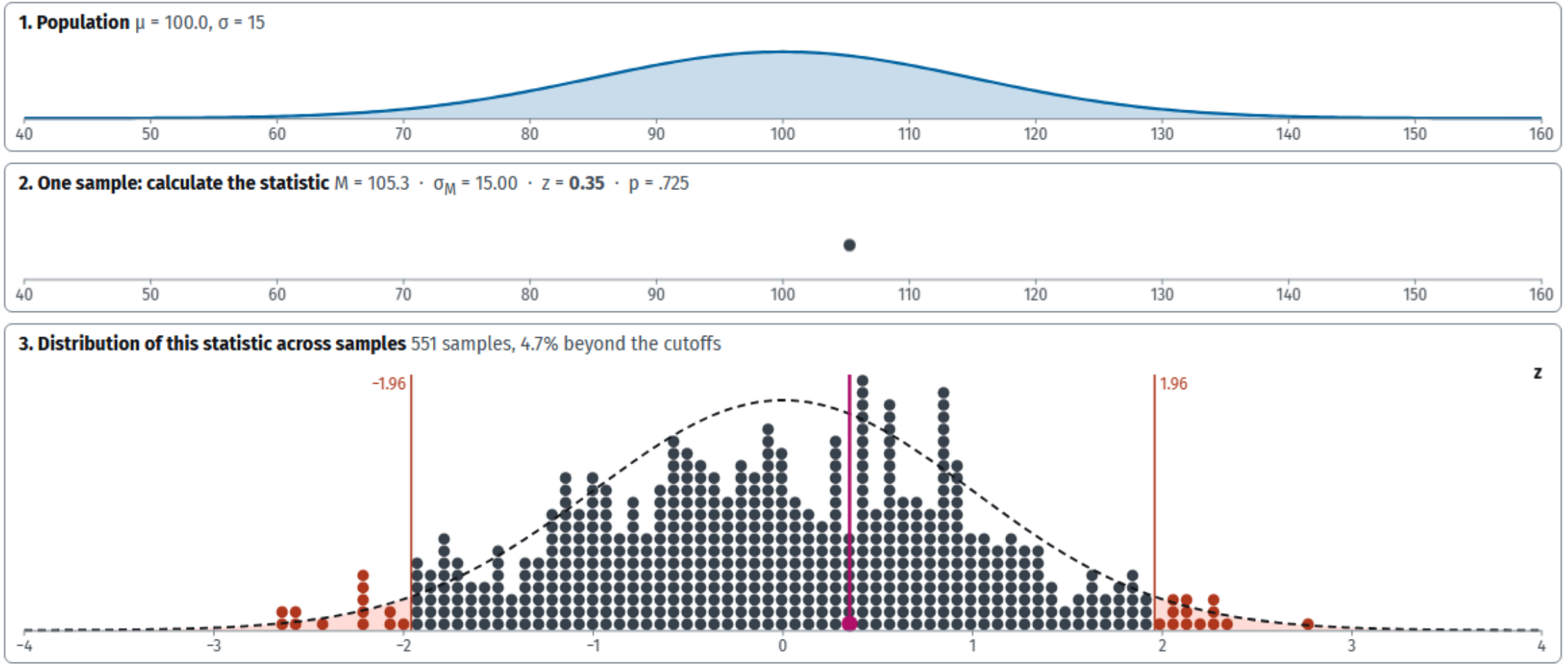
**Each dot is a whole sample, not a person**

# Count the dots to find a probability




357 of 551 means are below 106.2. So p(M < 106.2) is about 0.65

# Turn every mean into a z-score




Simulator after Kristoffer Magnusson, rpsychologist.com.

# So we can skip the counting

Any normal distribution in z-scores is the standard normal, so the unit normal table applies

$$z = \frac{M - \mu}{\sigma} = \frac{106.2 - 100}{15} = 0.41$$

Table: the body at z = 0.41 is 0.6591

Counting gave 0.65

A mean like ours isn't weird. About 66% of samples fall below it

# Predict it: A lopsided population

**Now the population is skewed right. We draw samples of 30 and compute each mean**

**What shape will the means make in panel 3?**

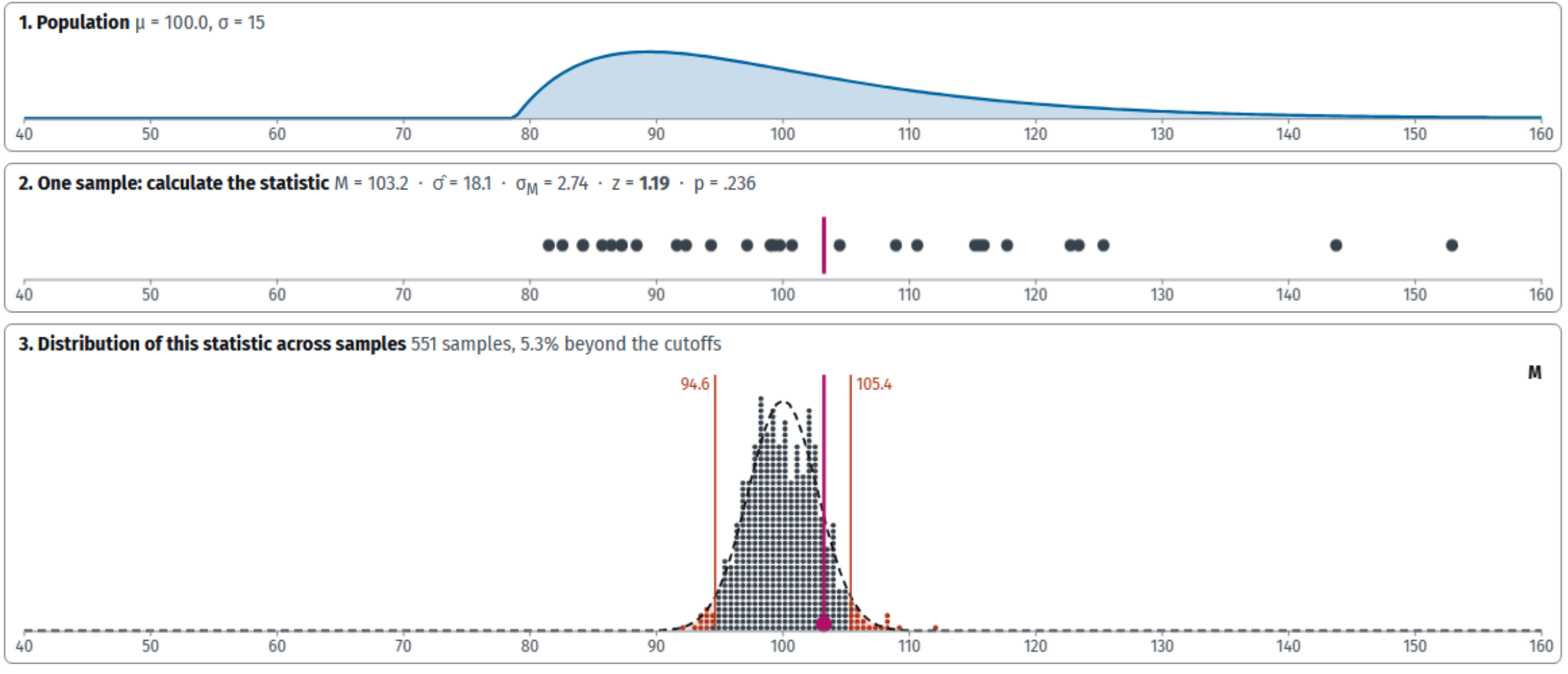
**A** Skewed right, like the population

**B** A bell around the population's mean

**C** Skewed left, the opposite way

**D** Flat

# Skewed population, bell-shaped means




Simulator after Kristoffer Magnusson, rpsychologist.com.

# The central limit theorem

**With large samples, the distribution of sample means is close to normal, whatever the population's shape**

If the population is normal, the means are normal at any n

If it isn't, n of about 30 is usually enough

So with a large n, the unit normal table works for means from any population

# Before Tuesday

**Friday, January 22: Lab 3 is a quiz. Bring a calculator**

**Sunday, January 24 at 6 PM: Mastery and Problem Set for Chapter 6**

**Finish Chapter 7 for Tuesday**

# Session 2: Tuesday, January 26

**Thursday in one line: means of samples pile up into a distribution of their own**

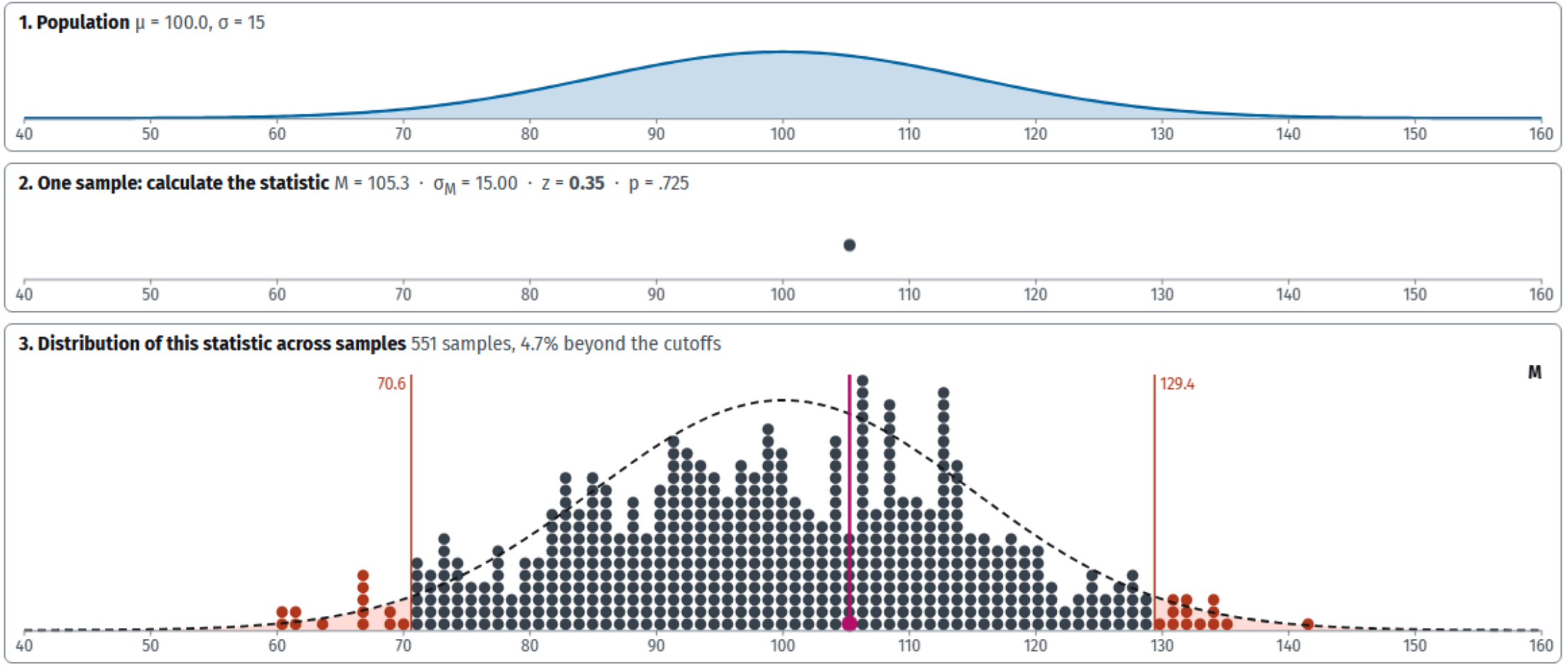
## Standard errors

## Estimating the standard error

## Using standard errors

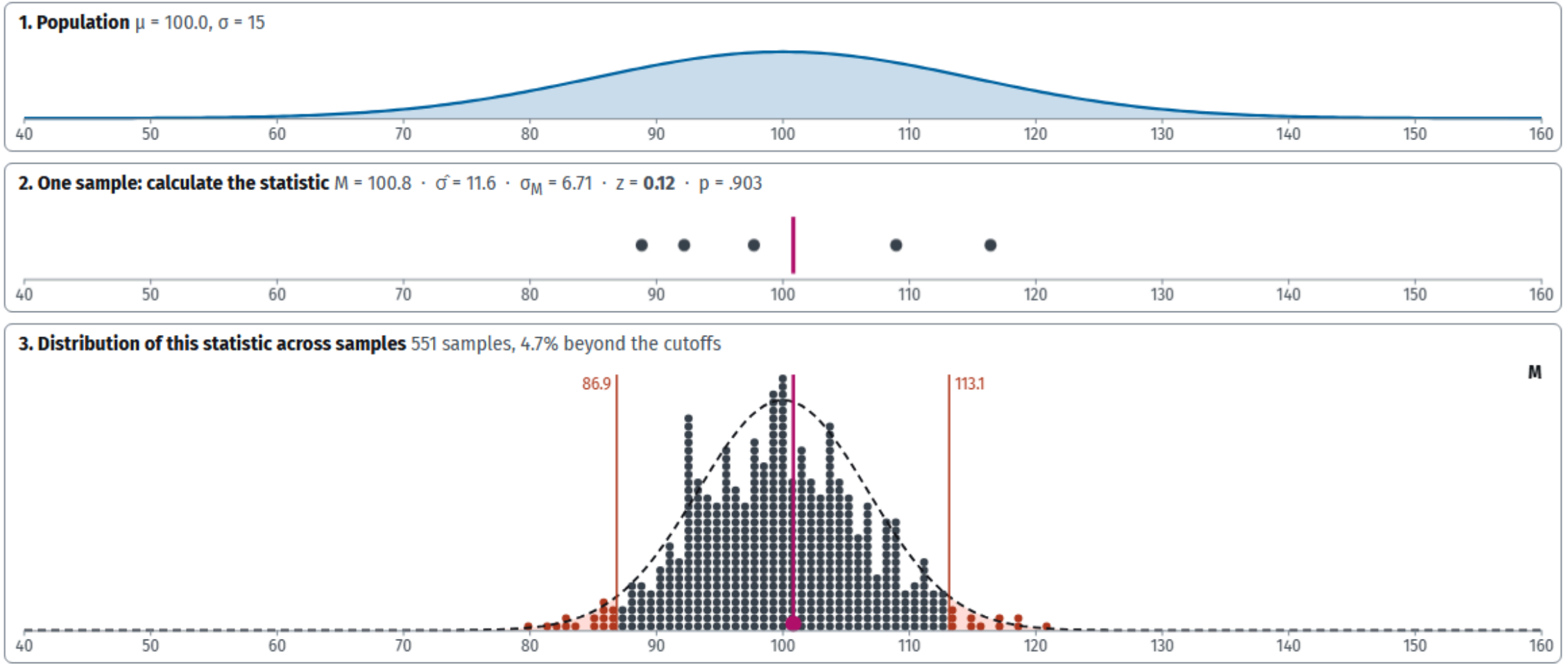
# Standard errors

# So far, each sample had one person



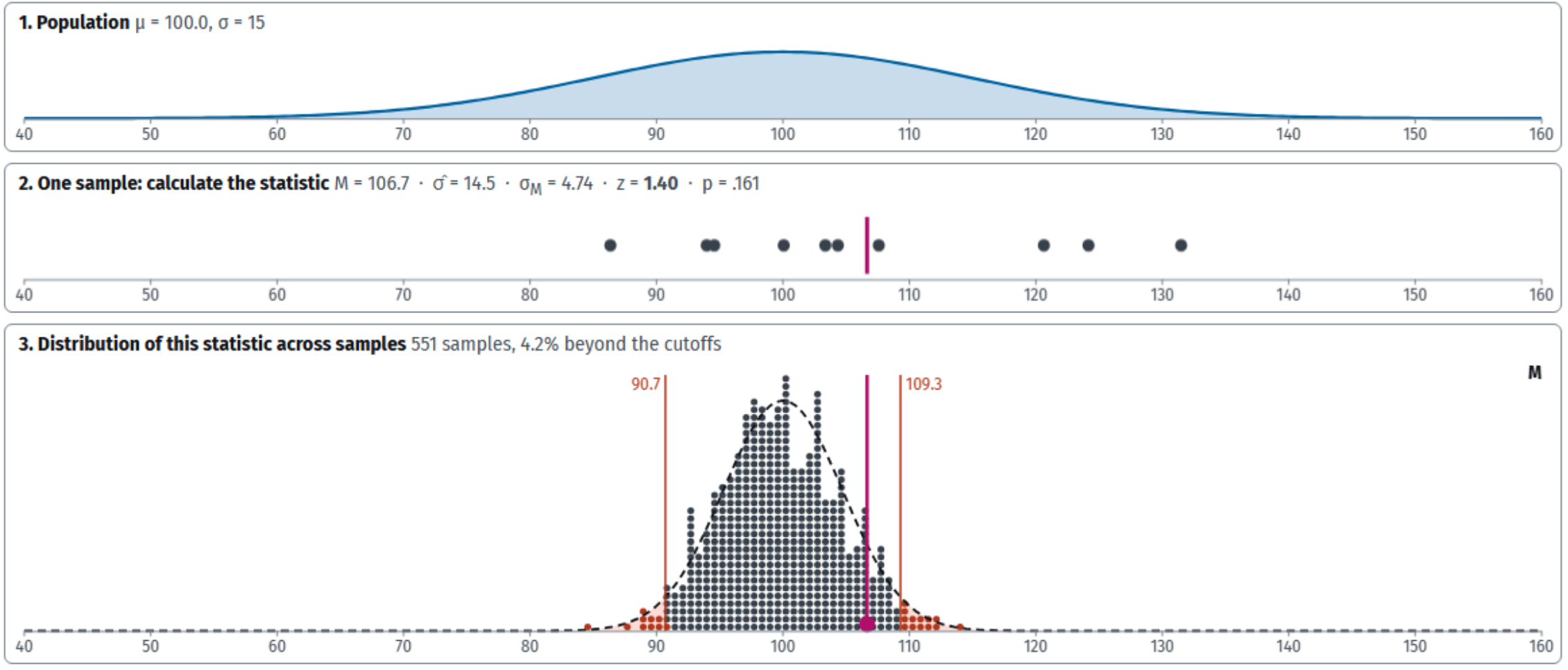

Simulator after Kristoffer Magnusson, rpsychologist.com.

# Now each sample has five people



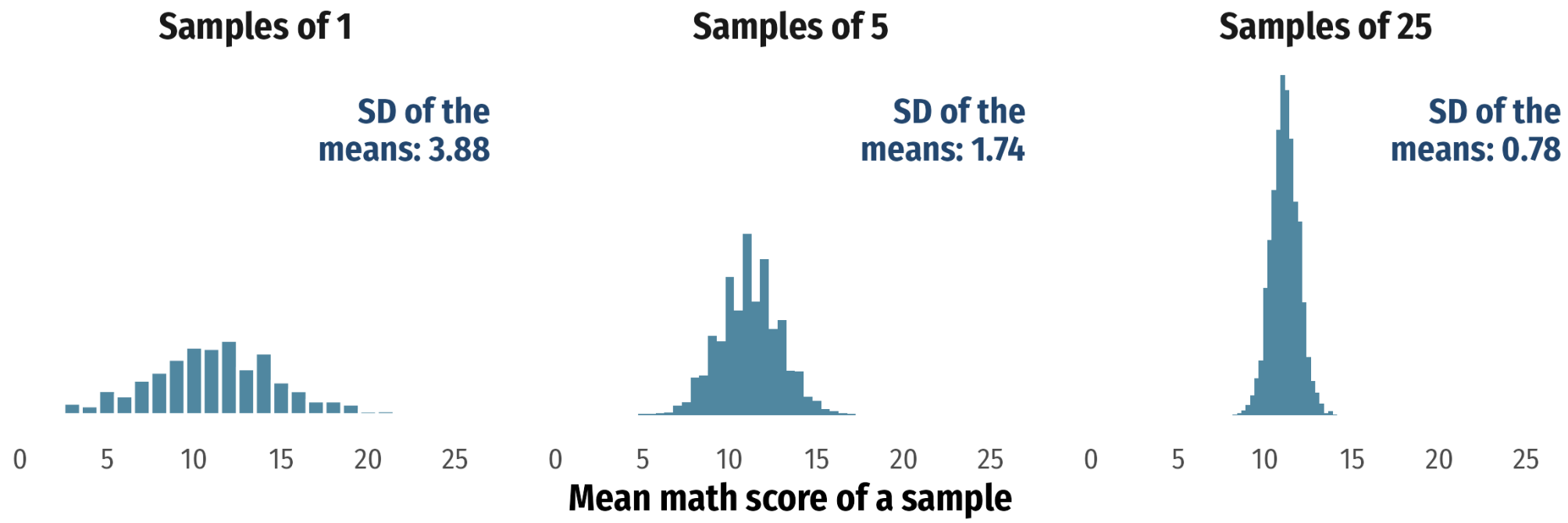

Simulator after Kristoffer Magnusson, rpsychologist.com.

# With ten, the means get pointier again




Simulator after Kristoffer Magnusson, rpsychologist.com.

# Our 619 students do the same thing




**The law of large numbers: the bigger the sample, the closer M tends to be to μ**

Data: Muralidharan, Singh, and Ganimian (2019). 5,000 samples of each size, treating the 619 students as a population.

# That spread is the standard error

$$\sigma_M = \text{the standard deviation of all the sample means}$$

**Sampling error**

The gap for one sample: M − μ

**Standard error**

The typical gap, across all samples

**With n = 1, $\sigma_M$ is just σ. As n grows, $\sigma_M$ shrinks**

# Predict it: How fast does it shrink?

Respond at **PollEv.com/adb**

**Same population, with σ = 15. With samples of 1, $\sigma_M$ = 15. Now each sample has 4 people. What's $\sigma_M$?**

**A** 15

**B** 7.5

**C** 3.75

**D** 0

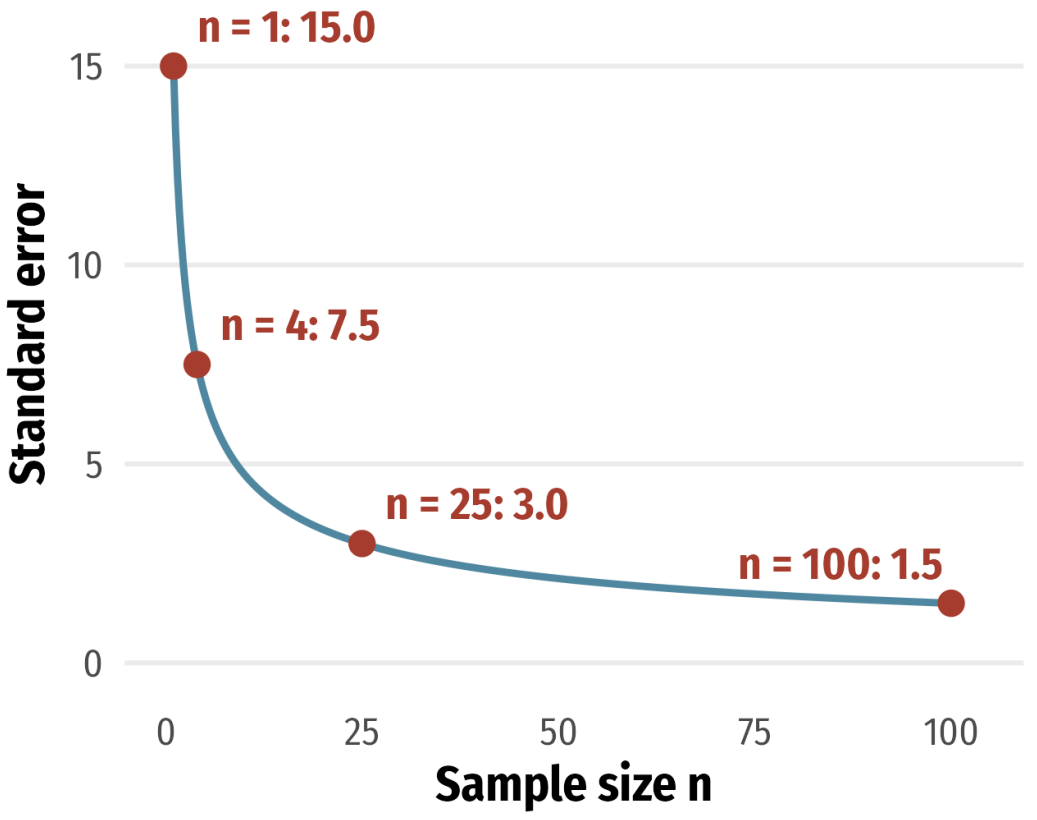
# Estimating the standard error

# Divide by the square root of n

$$\sigma_M = \frac{\sigma}{\sqrt{n}}$$

Our 619 students, n = 25: 3.90 / √25 = 0.78. The 5,000 samples gave 0.78

**To halve the standard error, you need four times the people**




Data: Muralidharan, Singh, and Ganimian (2019).

# The whole central limit theorem

## Shape

Close to normal as n grows, for any population

## Center

The mean of all sample means is $\mu$

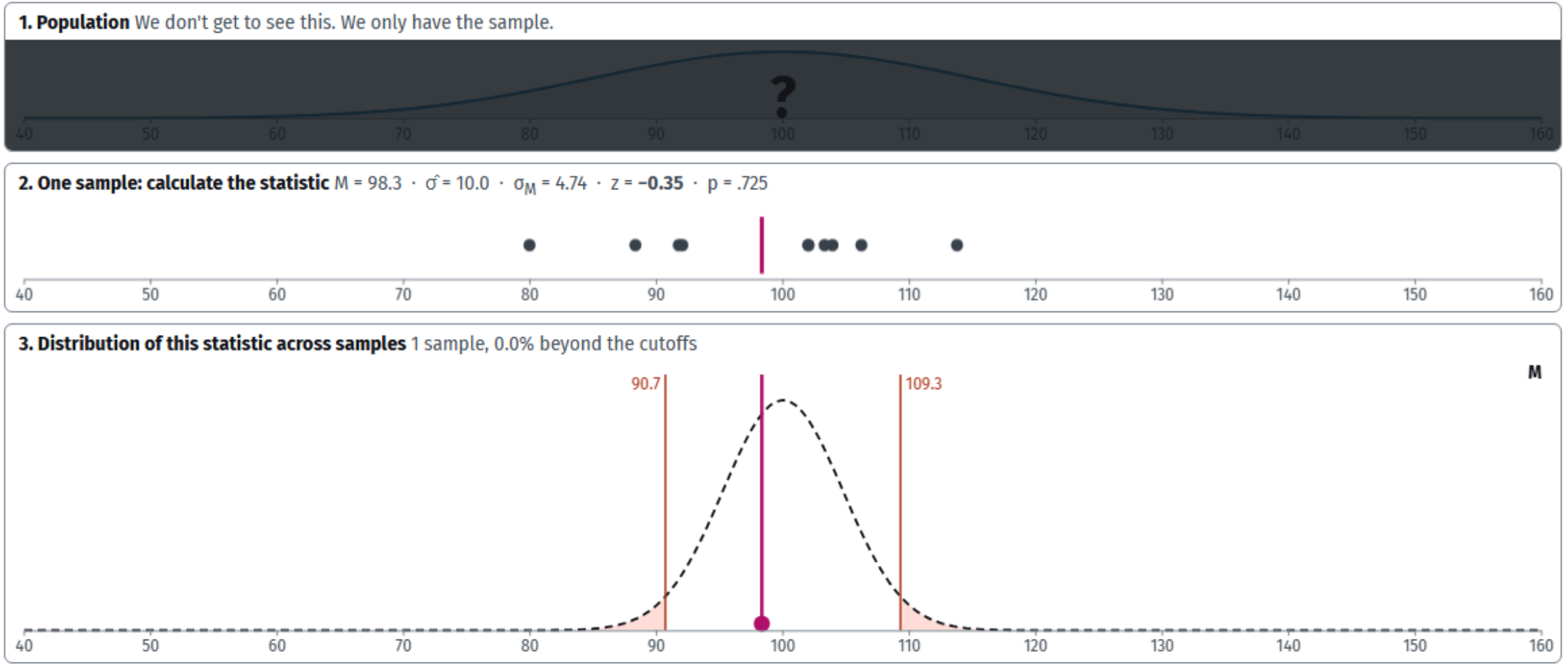
## Spread

The SD of all sample means is $\sigma / \sqrt{n}$

**Center and spread hold at any n. Only the shape needs a large n**

In the book: the mean of all sample means is the *expected value of M*, and $\sigma_M$ is the *standard error of M*

# But we only ever get one sample




Simulator after Kristoffer Magnusson, rpsychologist.com.

# So we estimate σ with our sample

$$\sigma_M = \frac{\sigma}{\sqrt{n}} \qquad \text{becomes} \qquad \hat{\sigma}_M = \frac{\hat{\sigma}}{\sqrt{n}}$$

**Greek: the true standard error. It needs σ**

**Hat: our estimate of it, from one sample**

In the book: $\hat{\sigma}$ is written s, and $\hat{\sigma}_M$ is written $s_M$. Chapter 9 introduces $s_M$

# From scores to the standard error

| Step | In symbols |
|---|---|
| 1. Find each deviation from the mean | $X - M$ |
| 2. Square them and add them up | $SS = \sum(X - M)^2$ |
| 3. Divide by n - 1 | $\hat{\sigma}^2 = SS/(n-1)$ |
| 4. Take the root | $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ |
| 5. Divide by the root of n | $\hat{\sigma}_M = \hat{\sigma}/\sqrt{n}$ |

**Steps 1 to 4 are Topic 3. Step 5 is the only new one**

# The same thing, written other ways

$$\hat{\sigma}_M = \frac{\hat{\sigma}}{\sqrt{n}} = \frac{\sqrt{\hat{\sigma}^2}}{\sqrt{n}} = \sqrt{\frac{\hat{\sigma}^2}{n}} = \sqrt{\frac{\text{variance}}{n}}$$

The last form: divide the variance by n, then take the root

That form comes back in Topic 9, when we compare two means

# One real sample of 25

| | The truth (all 619) | Our one sample |
|---|---|---|
| Mean | $\mu = 11.12$ | $M = 11.80$ |
| Standard deviation | $\sigma = 3.90$ | $\hat{\sigma} = 3.72$ |
| Standard error | $\sigma_M = 0.78$ | $\hat{\sigma}_M = 0.74$ |

**One sample gives an estimate, and a sense of how far off it may be**

Data: Muralidharan, Singh, and Ganimian (2019). One random sample of 25 from the 619 baseline scores.

# Try it: Four real students

## Math scores: 10, 8, 13, 13

**Find M, SS, $\hat{\sigma}$ and $\hat{\sigma}_M$**

**Work alone for two minutes. Compare with a neighbor. Then we vote**

Data: Muralidharan, Singh, and Ganimian (2019).

# Try it: Answer

$M = (10 + 8 + 13 + 13) / 4 = 11$

$\hat{\sigma} = \sqrt{(18 / 3)} = \sqrt{6} = 2.45$

$SS = (-1)^2 + (-3)^2 + (2)^2 + (2)^2 = 18$

$\hat{\sigma}_M = 2.45 / \sqrt{4} = 1.22$

**With four people, the mean could easily be off by a point or more**

Data: Muralidharan, Singh, and Ganimian (2019).

# Using standard errors

# A standard error does two jobs

## Job 1: Uncertainty

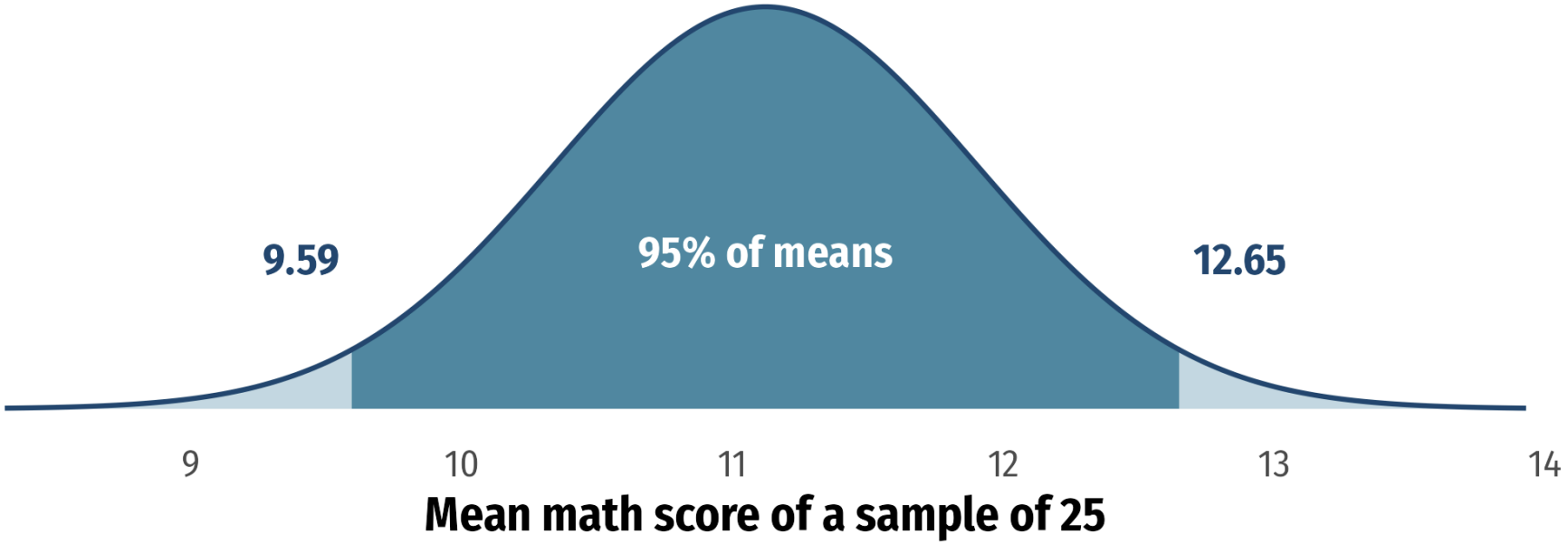
How far off might this estimate be?

## Job 2: Surprise

If I'm right about the population, how weird is this mean?

**The first job leads to confidence intervals. The second leads to hypothesis tests**

# Job 1: Where do means land?




$\mu \pm 1.96\ \sigma_M = 11.12 \pm 1.96 \times 0.78$. So 95% of samples land within 1.53 points of $\mu$

Data: Muralidharan, Singh, and Ganimian (2019).

# Polls call this the margin of error

## A poll of 1,000 people: 52% say yes, ± 3 points

The ± is 1.96 standard errors

Here, 1.96 × 1.58 = 3.1 points

Sample size drives it, not population size

Four times the people: half the margin

# Job 2: How surprising is this mean?

Locate a mean in its own sampling distribution, the way a z-score locates a score

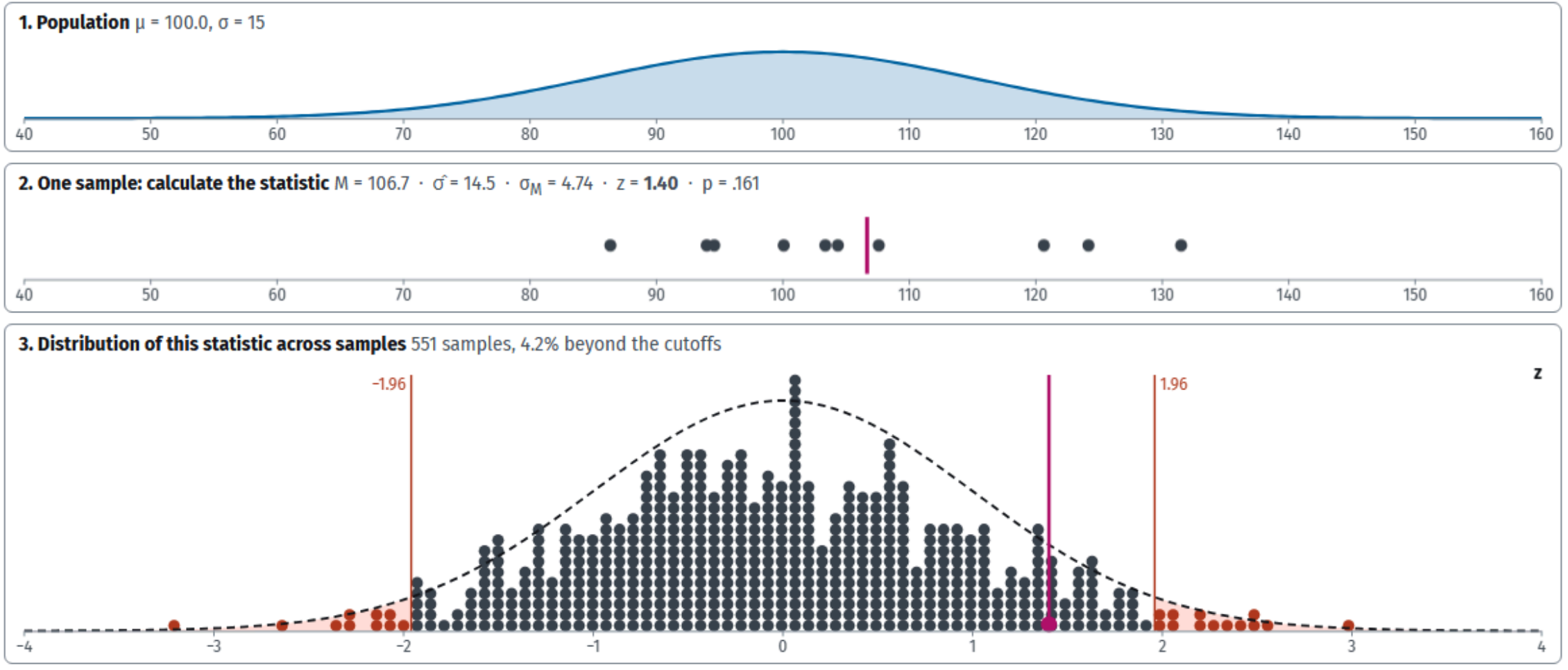
$$z = \frac{X - \mu}{\sigma} \quad \text{becomes} \quad z = \frac{M - \mu}{\sigma_M}$$

Top: how far M is from μ

Bottom: how far a mean typically falls

With n = 1, the two formulas are the same

# In z-scores, every n lands on one curve




Simulator after Kristoffer Magnusson, rpsychologist.com.

# Job 2 on real data: was the lottery fair?

The lottery picked 314 of the 619 students at random. So the winners are a random sample of our population

Baseline: $M = 10.984$, $\mu = 11.121$

$\sigma_M = 3.901 / \sqrt{314} = 0.220$

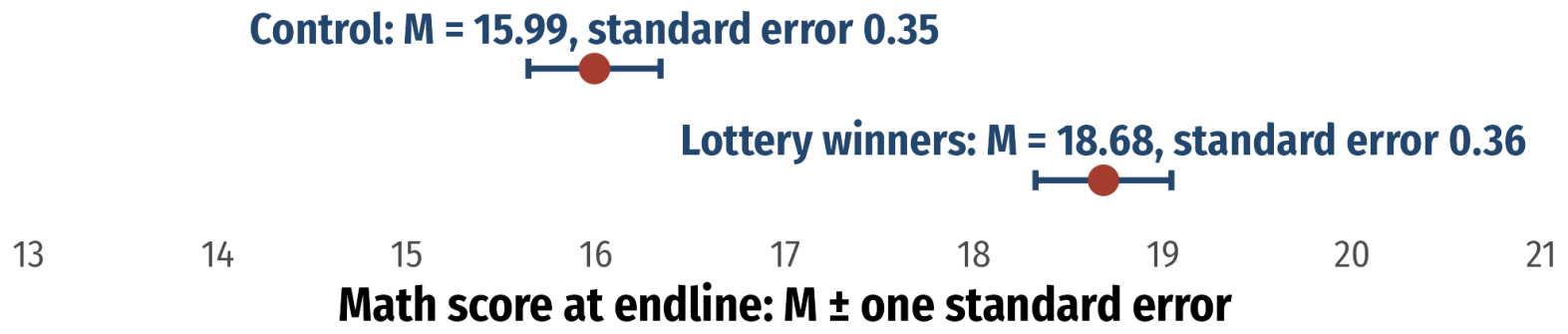
$z = (10.984 - 11.121) / 0.220 = -0.62$

53% of samples land this far from $\mu$ or farther

That's not weird at all, which is what a fair lottery should give

Data: Muralidharan, Singh, and Ganimian (2019).

# Predict it: After the program




**Could a gap like this be plain sampling error?**

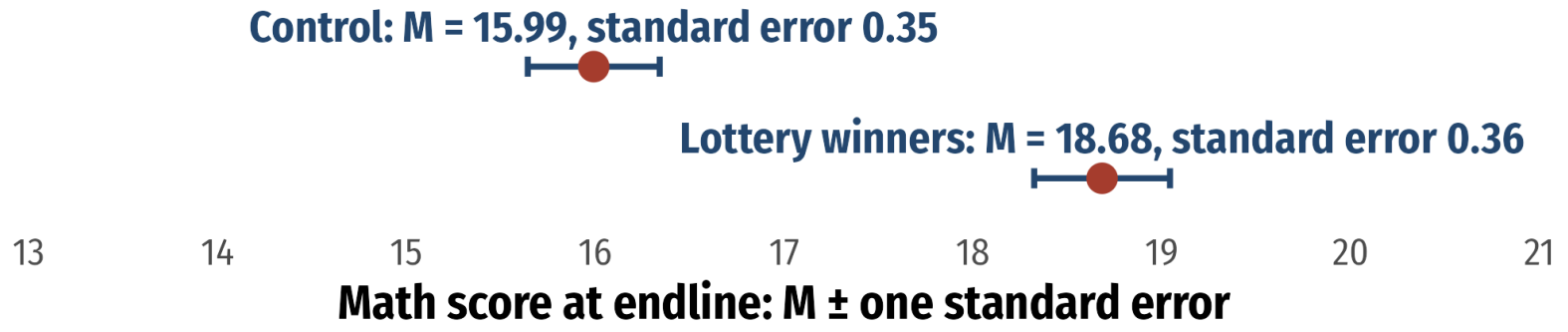
**A** Easily: both could come from one population

**B** Hardly: the gap is many standard errors

**C** No way to say without μ

Data: Muralidharan, Singh, and Ganimian (2019).

# The gap dwarfs the sampling error




**The gap is 2.69 points. Each mean is typically off by about 0.35 points**

**The answer is B. Topic 9 turns this into a real test, with a standard error for the gap itself**

Data: Muralidharan, Singh, and Ganimian (2019).

# Where are we?

**Description**

Summarize data

Visualize data

**Prediction**

**We are moving here**

**Explanation**

# What you can do now

Read the blueprint: population, one sample, distribution of a statistic

Say what a sampling distribution is, and what each dot in it stands for

State the central limit theorem: shape, center and spread

Compute $\sigma_M$ from $\sigma$, and $\hat{\sigma}_M$ from a sample

Use a standard error to find where means land, and a z-score for a mean

# Next steps

**Before Thursday**

Read Chapter 8. Topic 7 is hypothesis testing

**This Friday: Lab 4**

Sampling distributions and the four steps. Bring a calculator and a laptop

**Mastery and Problem Set for Chapter 7 are due Sunday, January 31 at 6 PM**